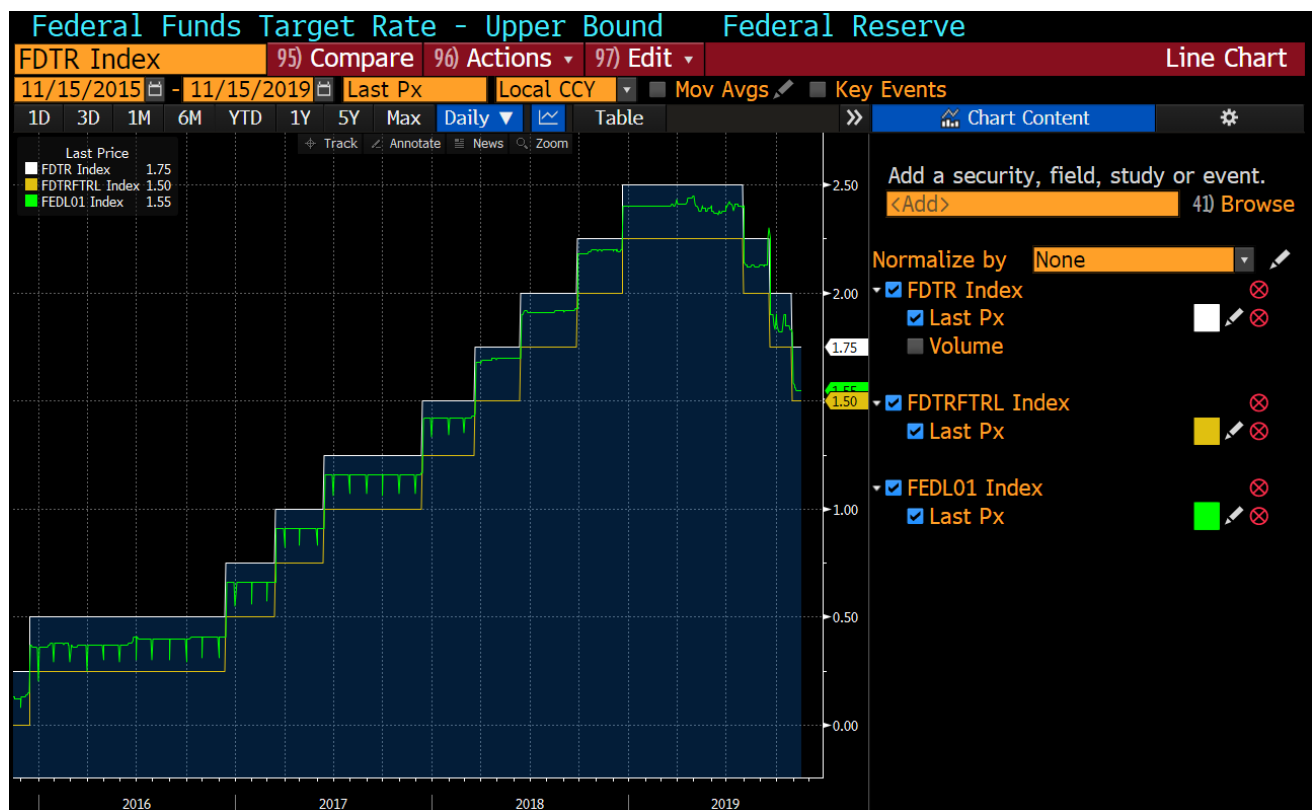


Calculations

This document provides examples of how WIRP leverages futures and OIS asset prices to calculate the number of hikes and cuts expected by a Central Bank. You will also learn the assumptions WIRP relies on to make calculations, how WIRP generates forward rates (including the data used), and how WIRP uses these forward rates to calculate the implied data displayed in WIRP.

Model Assumptions

Both the Futures and the OIS model rely on the assumption that only a central bank action will impact the effective interest rate of an economy. This assumption allows the expected overnight rates to be pushed forward and backward through the chosen asset's tenor structure, as shown in the following examples.



Futures Model

To perform the calculations for this model, WIRP collects price data for a list of assets from the Futures market and uses this data to derive interest rate projections for each upcoming CB meeting. WIRP assumes that for the current asset prices to exist the rate projections will hold true, as shown in greater detail in the following example.

9/16/2019 Example

The Effective Fed Funds Rate (EFFR) is the interest rate that is targeted by the Federal Reserve's Federal Open Market Committee (FOMC) for its implementation of US monetary policy. Simply speaking, it is the unsecured overnight lending rate between Federal Reserve member banks and sets a floor for all interest rates prevailing in the United States. If you wanted to hedge an overnight rate exposure, you might purchase a [CME 30 Day Fed Funds Future](#). A future is expressed as $100 - R$ where R is the expected arithmetic average overnight rate for the contract month and where the overnight rate is the expected value of EFFR referenced above. For example, the Dec2019 contract, as of 9/16/2019, had a quoted price of 98.295. This implies a monthly average expected overnight rate of 1.705. The following table shows an example of how this might look.

Dec2019 Futures Contract Scenario 1																															
Date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Expected Effective Rate	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705	1.705
Month Average Rate	1.705																														

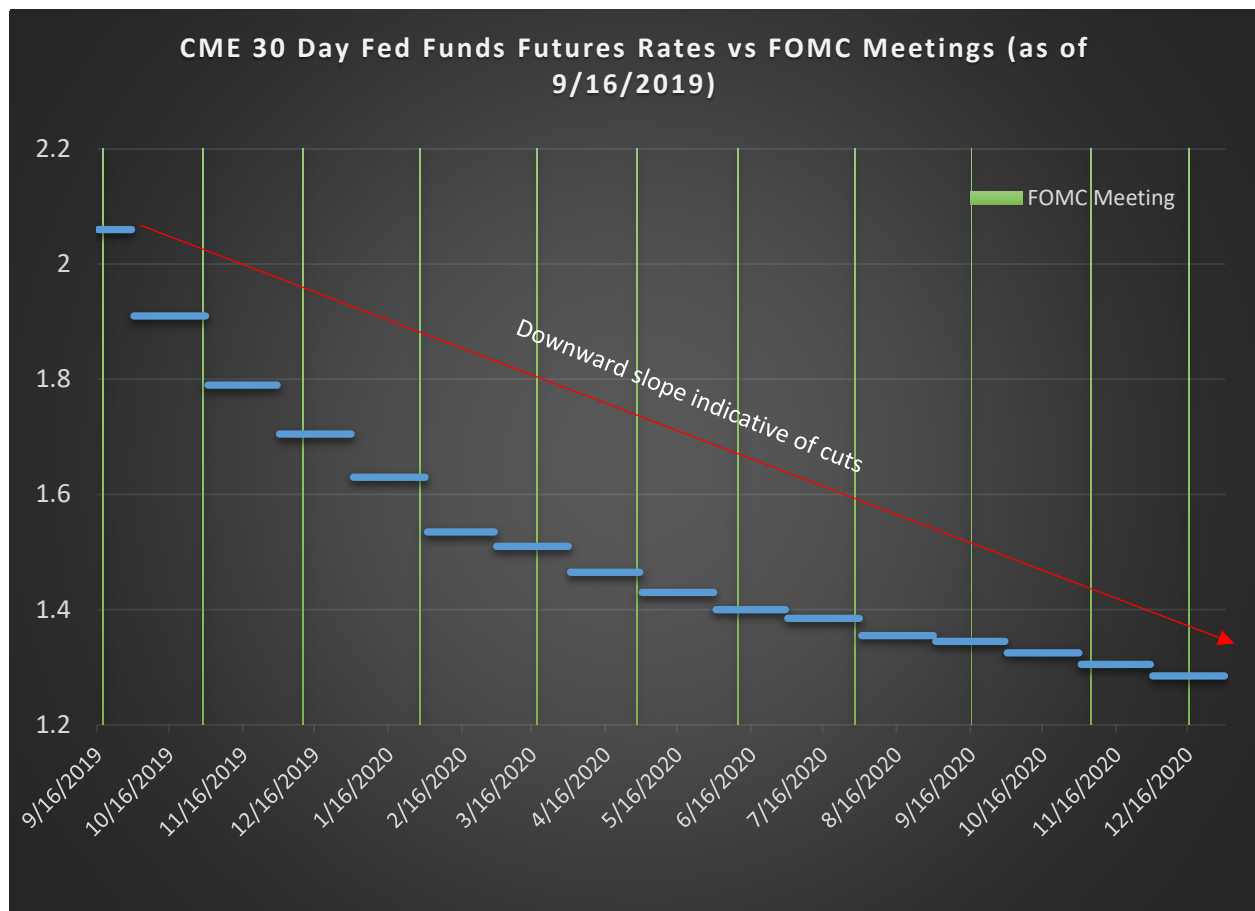
Legend

2	Business Days
1	Weekend Days
11	FOMC Meeting Days

The scenario above indicates that the rate is expected to be the same throughout the contract, even though there is an FOMC meeting expected on the 11th of the month. In reality, assuming a potential hike or cut, the contract would likely have one expected rate before the meeting and another expected rate after.

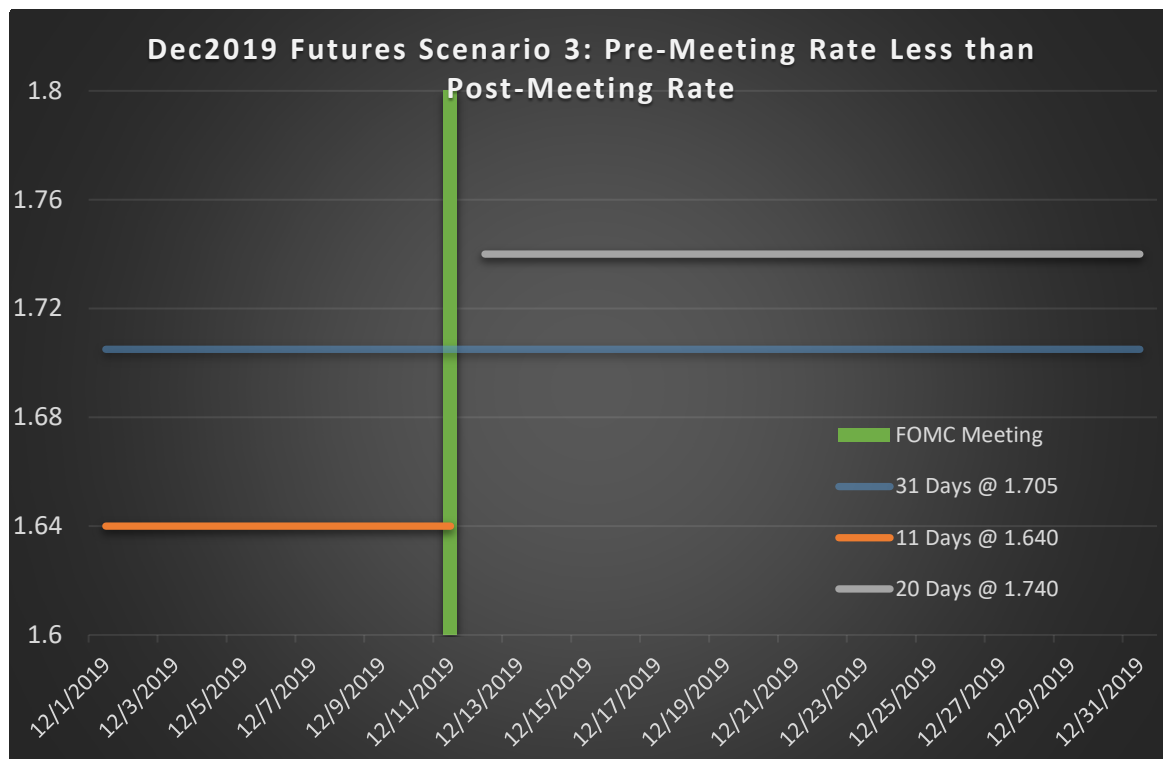
Note: To see these securities, run *FFA Comdty CT <GO>*, or click **Instrument** at the top of the WIRP screen.

Once we acquire the overnight rates for the futures contracts, we can start to make interest rate predictions. The following chart shows the eight Federal Open market Committee (FOMC) meetings held per year and the Federal Funds Futures (FFF) contracts for each month. The chart also indicates that the market expectation is that the FOMC will be cutting interest rates.

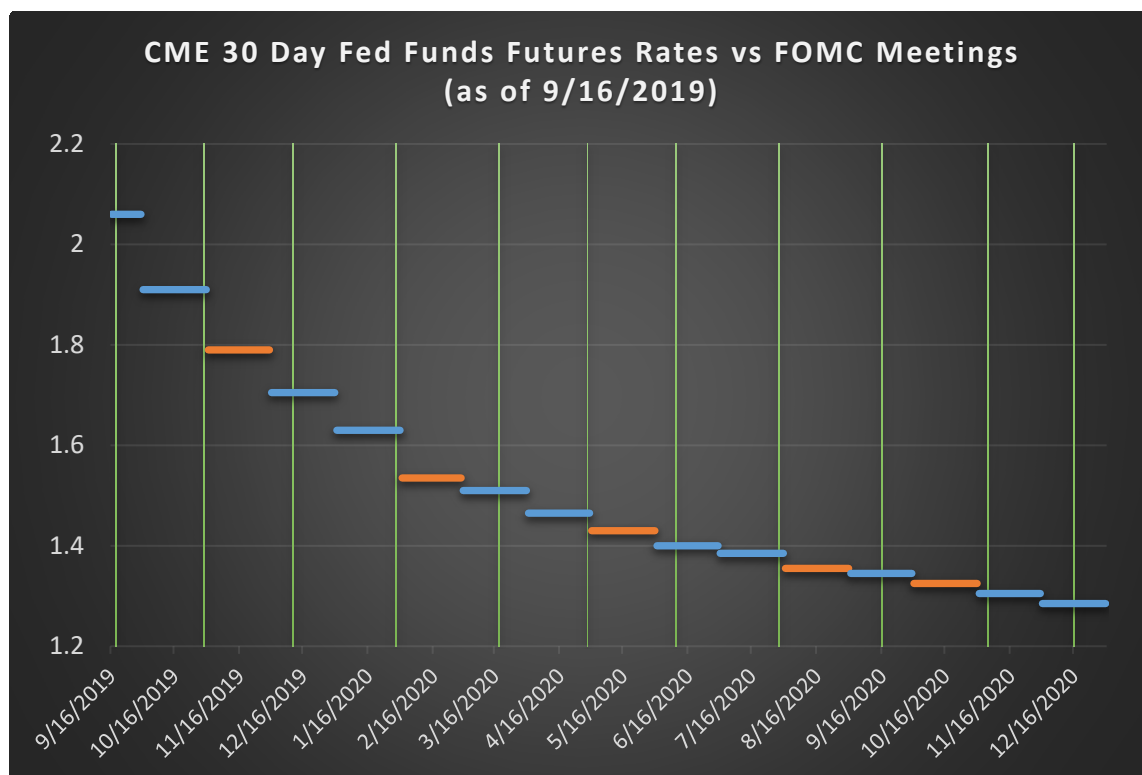


Focusing back on our December 2019 contract, we don't know the interest rates expected both before and after a meeting. For instance, our December 2019 rate of 1.705 could imply a higher rate prior to the meeting and a lower rate following, as shown in the following scenario.

Dec2019 Futures Contract Scenario 2																															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Price	1.78	1.78	1.78	1.78	1.78	1.78	1.78	1.78	1.78	1.78	1.78	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	1.664	
Implied Vol	1.705																														

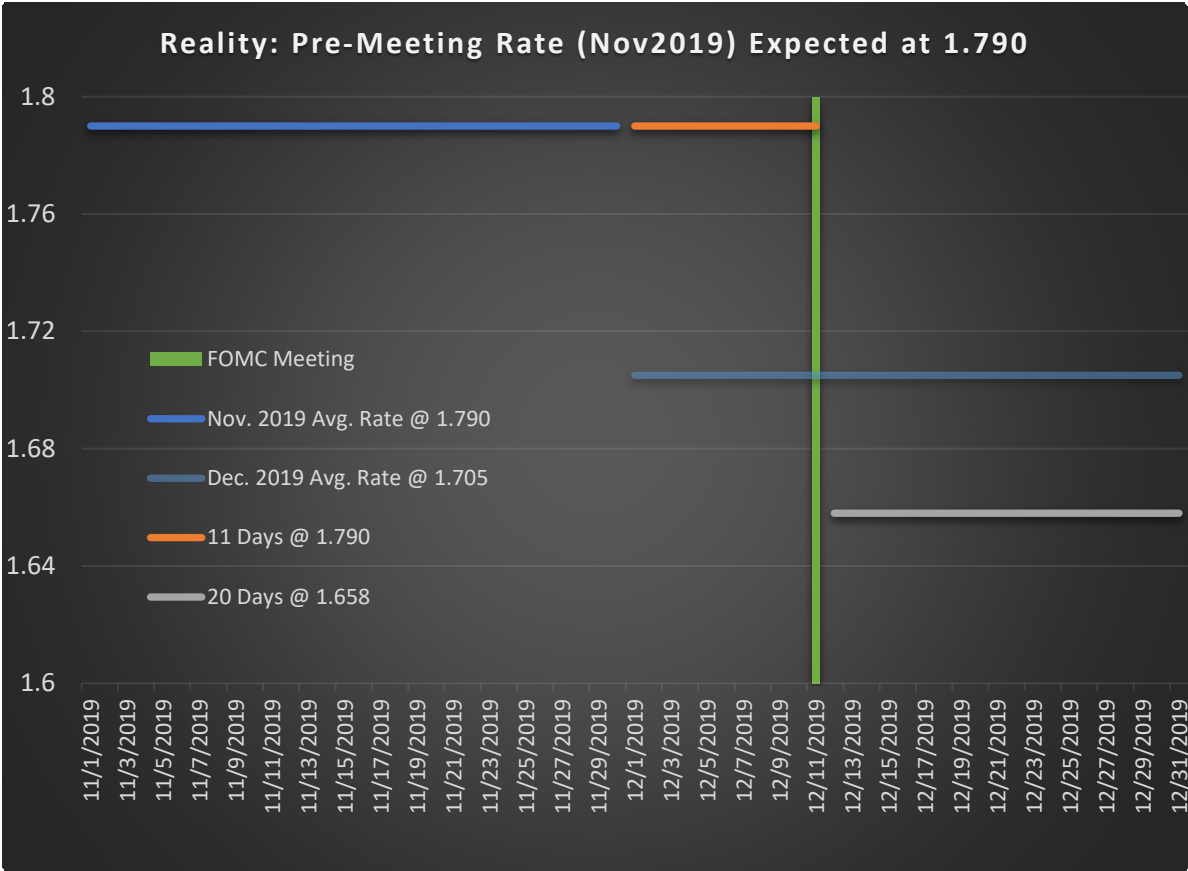


Taken individually, each contract that spanned a meeting month could be expressed as an equation such as, $31(1.705) = 11(y) + 20(x)$. Unfortunately, this is one equation with two unknowns. However, taken collectively we can see that there are contract months which do not span CB meetings.

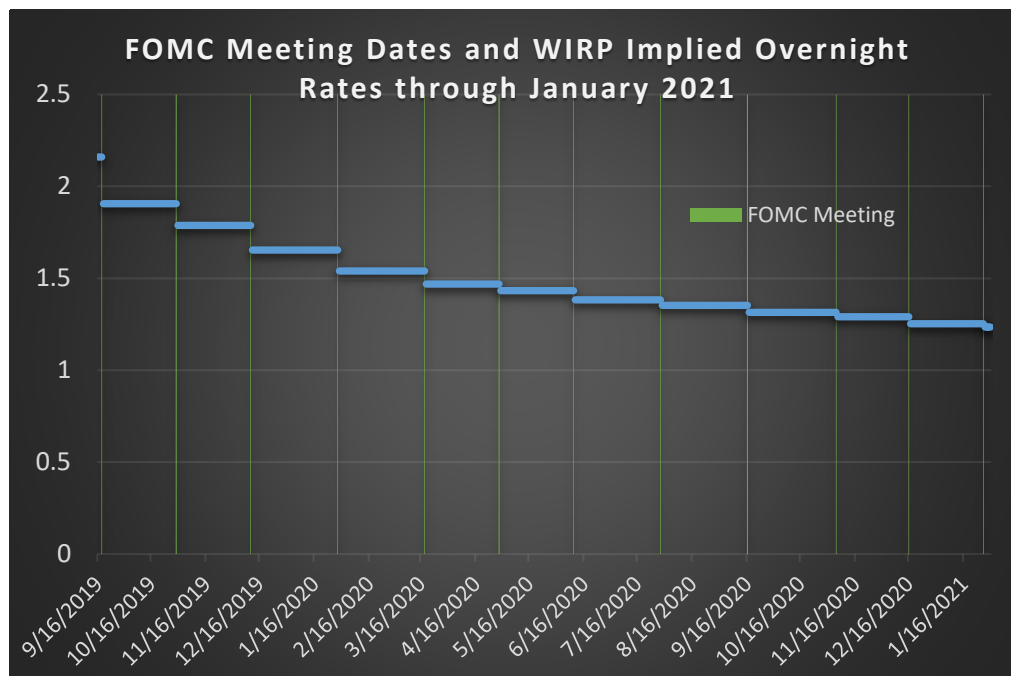


Based on the assumption that an action by a Central Bank is the only thing that should noticeably impact the effective rate of a given economy and that no meeting is expected in the November 2019 contract month, we can carry the November rate forward into the Dec2019 contract. Doing so results in an equation such as, $31(1.705) = 11(1.790) + 20(x)$. Now, we simply need to solve for x (our forward rate), which, in this case, is approximately equal to ~1.658.

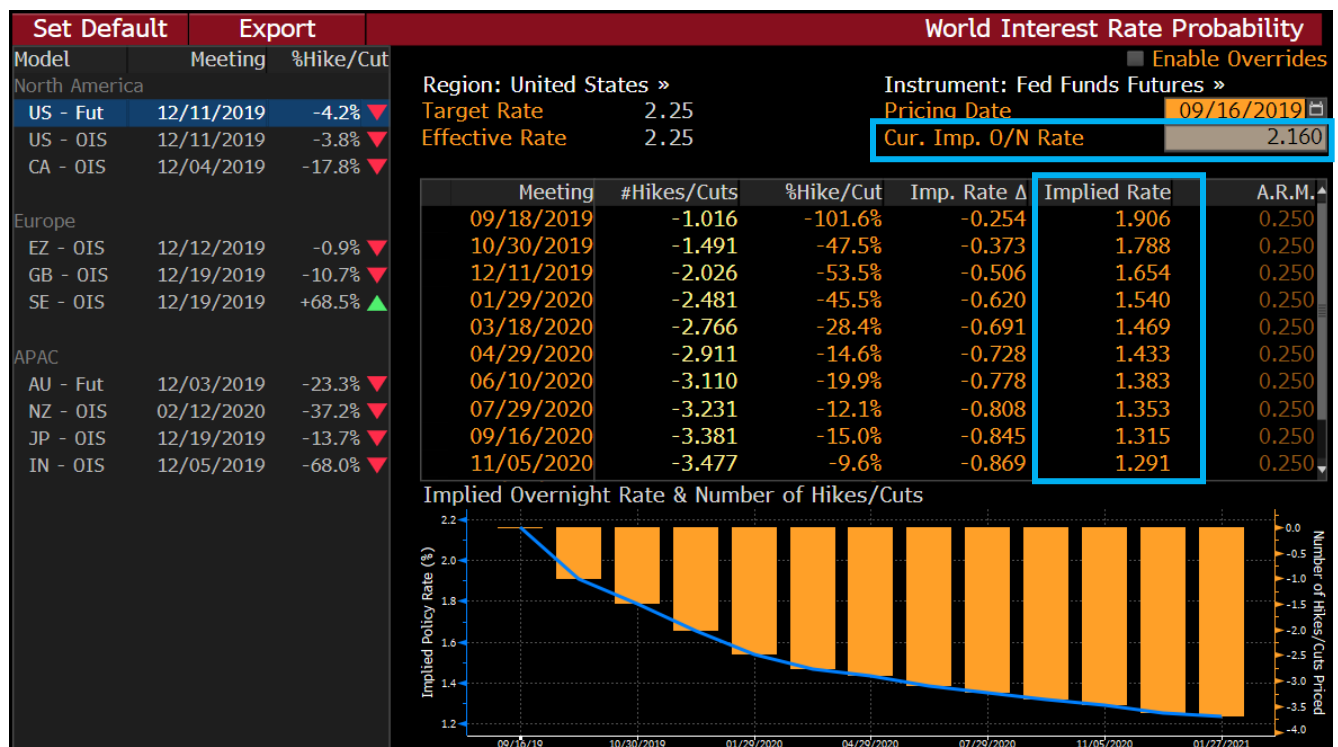
Dec2019 Futures Contract (9/16/2019)																															
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	
1.79	1.79	1.79	1.79	1.79	1.79	1.79	1.79	1.79	1.79	1.79	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	1.658	
1.705																															



After solving for the post-meeting rate, implied by the November and December contracts, we would work our way through the available contract prices and CB meeting dates to solve for all available forward rates.



Finally, we would take the data from curve that we constructed and express it in one of two categories: the Current Implied Overnight Rate, and the Post-Meeting Implied Overnight Rate. In this example, the Current Implied O/N Rate, the overnight rate expected to prevail up until the first meeting, is ~2.160. You can see the Post-Meeting Implied Overnight Rates, the interest rates that are expected to follow each FOMC meeting, for each meeting in the *Implied Rate* column.



Once we have these implied overnight rates, additional calculations are performed to estimate the extended WIRP data. For more information about the additional calculations, see [Implied Data](#).

OIS Model

This methodology exploits the relationship between the fixed and floating sides of monthly Overnight Index Swaps (OIS) to imply the rates surrounding CB meeting dates.

For this calculation, WIRP assumes that the following equation must be in balance for the market to be in equilibrium.

$$1 + \frac{r * n}{360} = \prod_{i=1}^n \left(1 + \frac{r_i * d_i}{360} \right)$$

Here the left side represents the fixed portion, and the right side represents the float. Where:

r = The fixed interest rate (quoted OIS Price)

r_i = The expected floating interest rate on the i th day

d_i = The number of days that r_i applies (1 for weekdays, 3 for weekends)

n = The total number of days for the swap

A fictional one-week example might look like:

$$1 + \frac{r * 7}{360} = \left(1 + \frac{r * 1}{360} \right)^2 \left(1 + \frac{r * 2}{360} \right) \left(1 + \frac{r * 3}{360} \right) \quad (1)$$

Where:

Term	Represents
$\left(1 + \frac{r * 1}{360} \right)^2$	Monday and Tuesday
$\left(1 + \frac{r * 2}{360} \right)$	Wednesday and Thursday assuming Thursday is a market holiday and the rate carries over.
$\left(1 + \frac{r * 3}{360} \right)$	Friday, Saturday, and Sunday

Given that r , n , and d_i are known, the only unknowns in the equation are the r s. As we have one equation with potentially many unknowns (the different r s) we cannot solve for this algebraically. In WIRP, we assume that these rates are the same and employ Newton's method to solve for a single

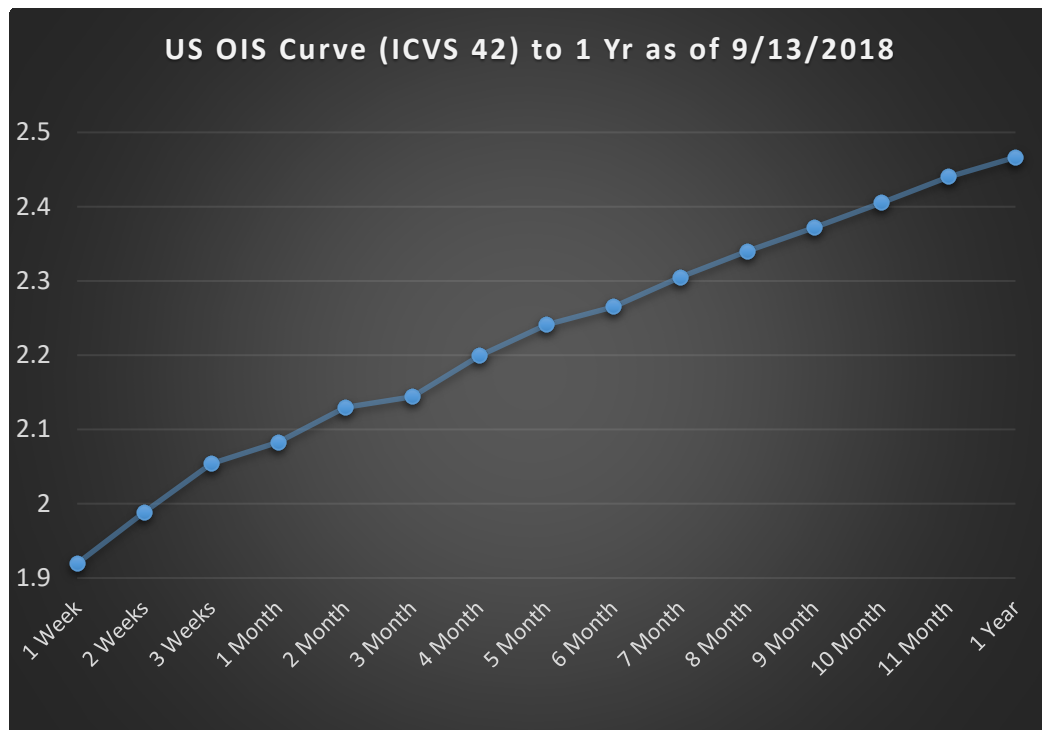
representative value of r , yielding what we refer to as the average expected variable rate over the period under scrutiny.

Using this assumption, we can solve for an average expected overnight rate associated with each tenor, paying specific attention to the tenors before and after Central Bank meetings. The following example from the United States illustrates this in more detail.

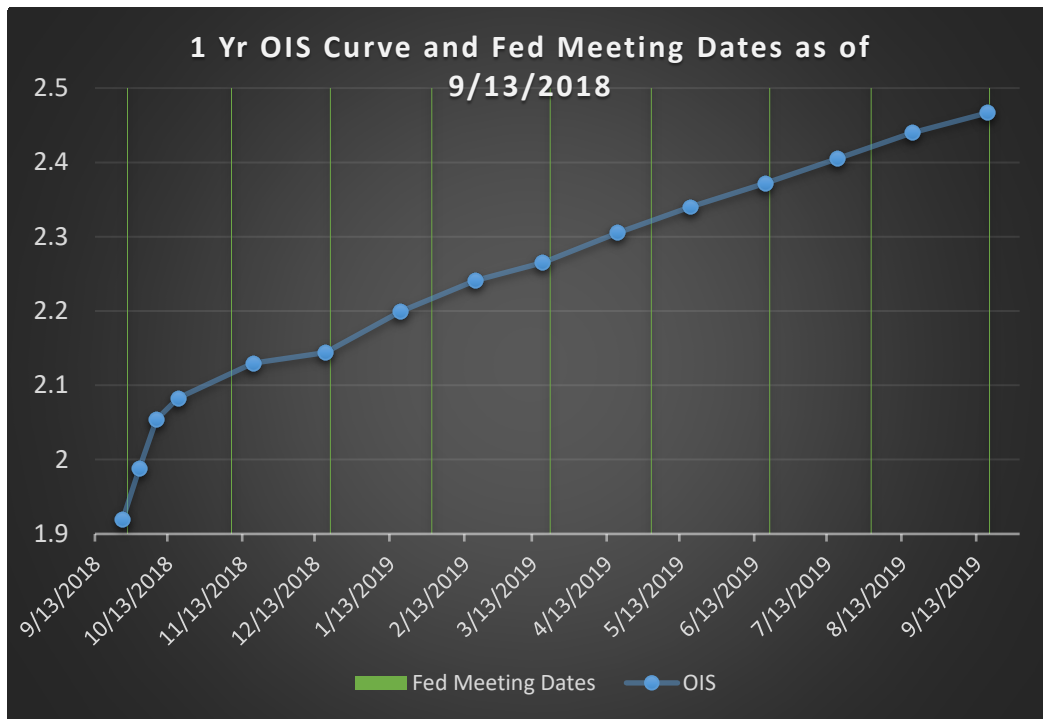
9/13/2018 Example

The following chart shows the US OIS curve as found at {ICVS 42<GO>}, as of 9/13/2018. The upward sloping curve indicates that the market expects that the Central Bank will be holding or hiking rates, not cutting.

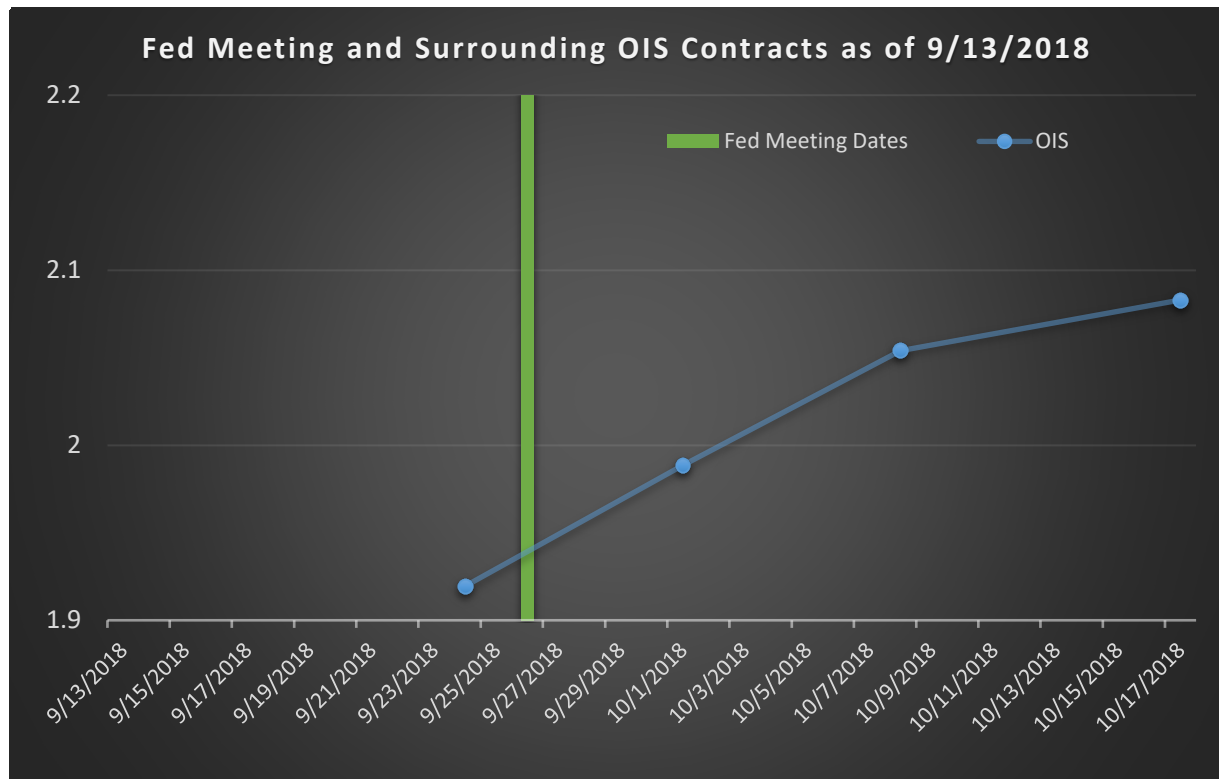
Note: To view the securities shown in the chart, run ICVS 42 <GO> or click **Instrument** at the top of the WIRP screen.



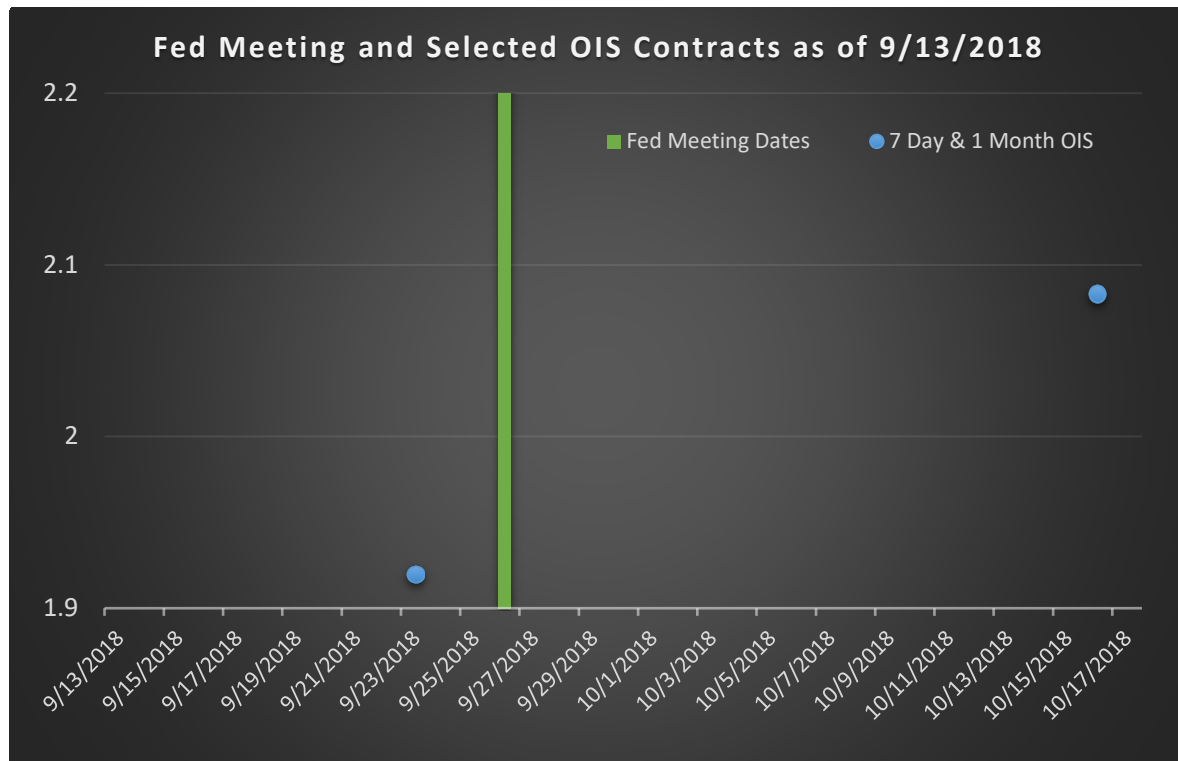
Solving for the settlement dates and injecting the Federal Reserve meeting dates would result in the following chart:



Zooming into the upcoming Fed meeting should result in the following chart:



While we could use the one-week and two-week securities for our calculations, we'll select the one week (to estimate the rate expected before the meeting) and one month (to estimate the rate expected after the meeting). In this example, we select the monthly tenor as it tends to have more liquidity than the similar three-week security. This leaves us with the following chart:



Calculating the Average Expected Rate before a Meeting

We can now begin making the calculations necessary to estimate the overnight rate expected by the market, both before and after the meeting of the Federal Reserve.

Note: Spot lag may create may create issues. Spot lag is the difference between when the contract is entered into and the first date of the floating side of the equation or fixing. While spot lag varies from market to market, it is generally regarded as two days in the OIS market. This is the convention followed in WIRP. If this lag lands the start of the contract on a weekend, accrual begins on the following business day. These dates can be referenced on the Bloomberg Swap Manager at SWPM –OIS R09/13/2018 <GO>.

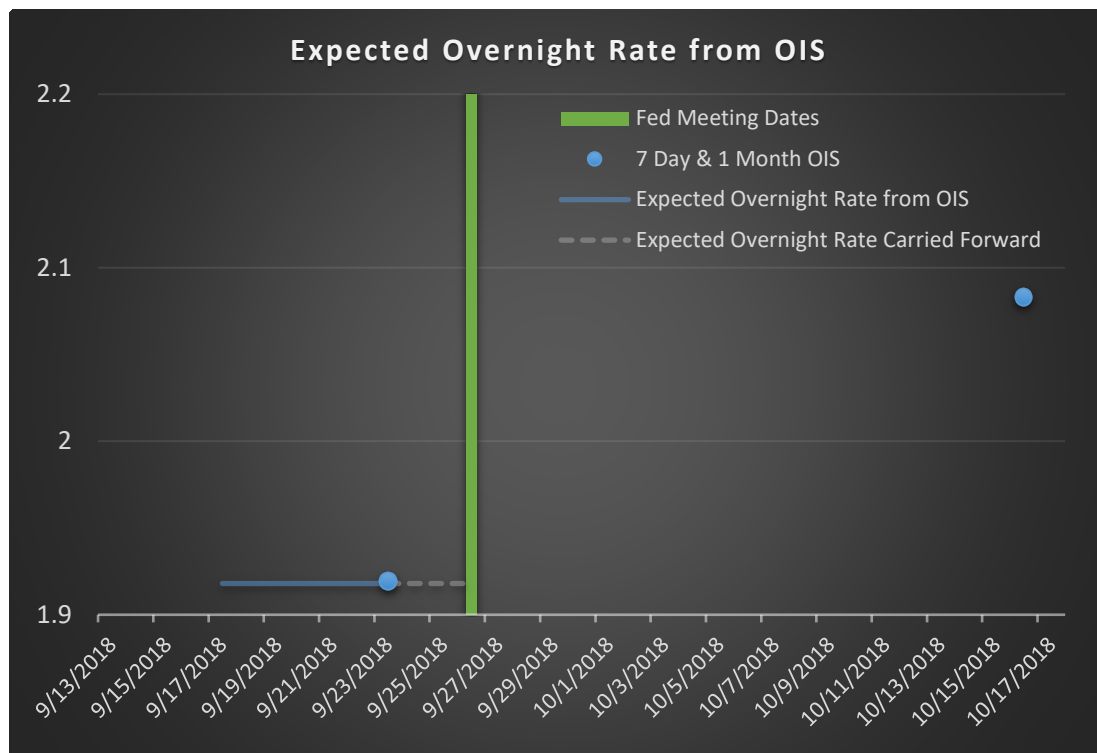
The quoted BGN price of a one-week OIS contract as of Sept. 13 was 1.9195. (For the purposes of fixing, accrual will begin on the 17th.) A two-day delay puts the start day on Saturday the 15th which implies that interest accrual starts on Monday the 17th and continues through the 23rd. Knowing that this period has four trading days and three weekend days, our equation now looks like:

$$1 + \frac{0.01920 * 7}{360} = \left(1 + \frac{r * 1}{360}\right)^4 \left(1 + \frac{r * 3}{360}\right) \quad (2)$$

Where the first term on the right hand side represents the weekdays, and the second represents the weekend days. We assume that all unknown r s are the same and solve for this 'average expected' value via Newton's method. On the close of business on the 13th, $r \approx 0.01918$, so we can say that from the 13th through the 23rd of September the expected overnight rate should be 0.01918 as expressed in equation 4.

$$1 + \frac{0.01920 * 7}{360} = \left(1 + \frac{0.01918 * 1}{360}\right)^4 \left(1 + \frac{0.01918 * 3}{360}\right) \quad (3)$$

Again, invoking our assumption that only a Central Bank Meeting decision should impact expected overnight rates, we carry this rate forward until the meeting.



Once we have this calculated value, we can calculate the implied forward rate expected after the meeting.

Calculating the Average Expected Rate after a Meeting

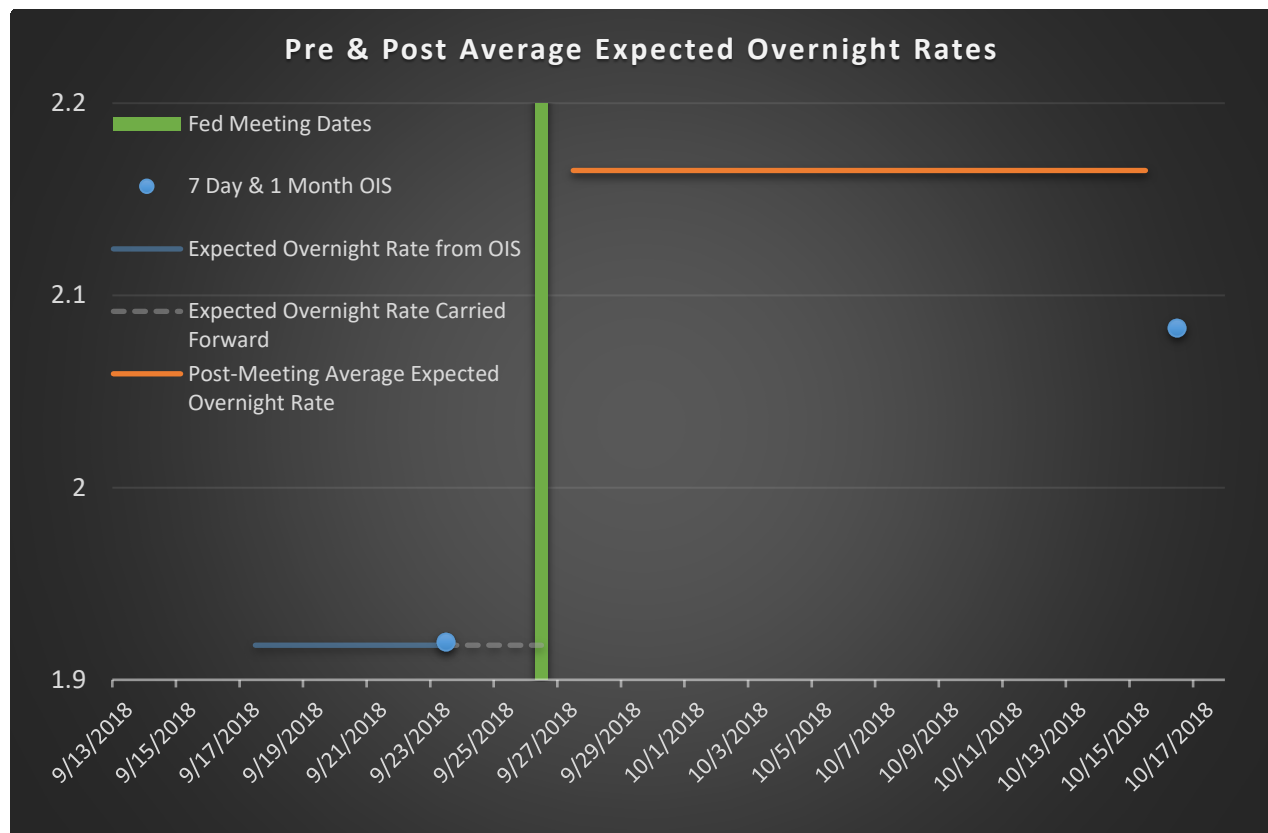
The quoted BGN price of a one month OIS contract as of Sept. 13 was 2.0830. Given spot lag, interest accrual begins on the 17th and continues through Oct. 16th. We can now fit this information into our equation along with the information we calculated above, as shown in the following equation.

$$\begin{aligned}
 & 1 + \frac{0.02083 * 30}{360} \\
 = & \left(1 + \frac{0.01918 * 1}{360}\right)^4 \left(1 + \frac{0.01918 * 3}{360}\right) \left(1 + \frac{0.01918 * 1}{360}\right)^3 \\
 & \left(1 + \frac{r * 1}{360}\right) \left(1 + \frac{r * 3}{360}\right) \left(1 + \frac{r * 1}{360}\right)^4 \left(1 + \frac{r * 4}{360}\right) \left(1 + \frac{r * 1}{360}\right)^3 \\
 & \left(1 + \frac{r * 3}{360}\right) \left(1 + \frac{r * 1}{360}\right)^2
 \end{aligned} \tag{4}$$

Note that we substitute in the average expected rate from our previous calculation for the r of the first two terms and include the same rate for the third term (the period between the end of accrual on the one-week security and the Fed Meeting). The following table describes the terms on the float side of the equation with our 'average expected' r replacing the r_i s from equation 1.

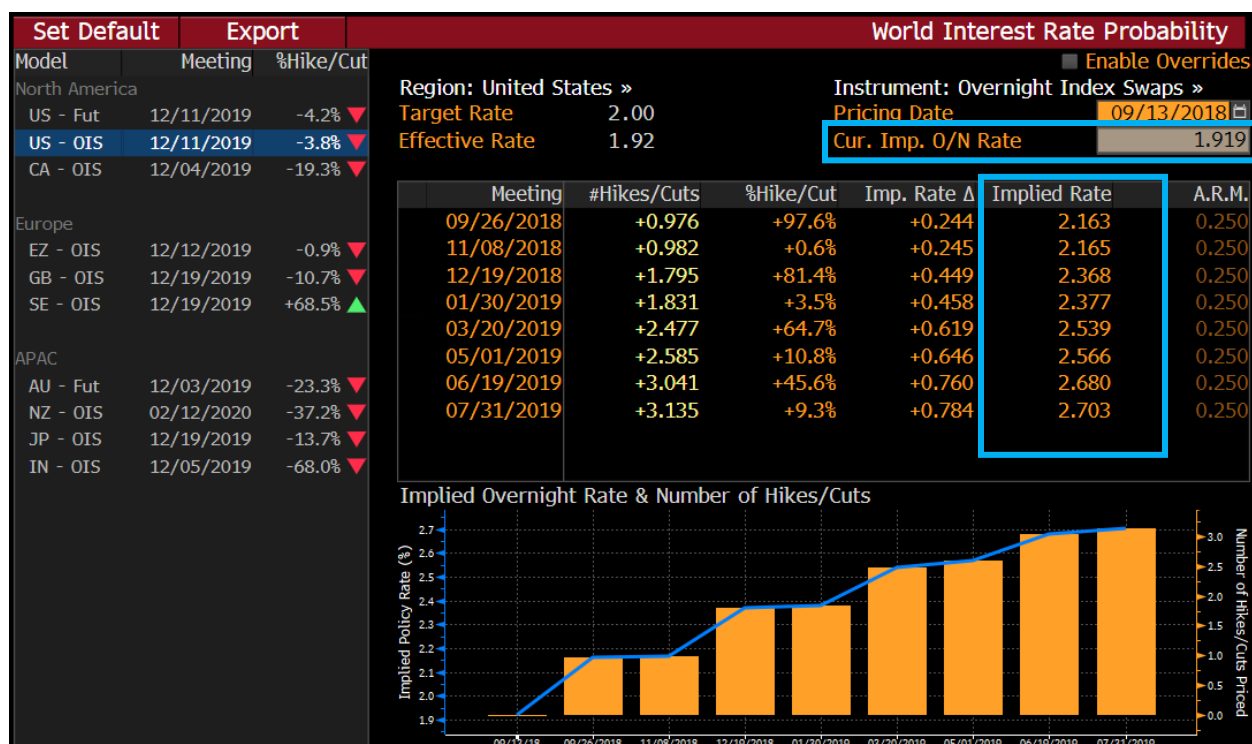
9/17 - 9/20	Weekdays (same as Eq 4)	$\left(1 + \frac{0.01918 * 1}{360}\right)^4$
9/21 - 9/23	Weekend Days (same as Eq 4)	$\left(1 + \frac{0.01918 * 3}{360}\right)$
9/24 - 9/26	Weekdays Through Meeting (Implied by Eq 4)	$\left(1 + \frac{0.01918 * 1}{360}\right)^3$
9/27	Weekday After Meeting	$\left(1 + \frac{r * 1}{360}\right)$
9/28 - 9/30	Weekend Days	$\left(1 + \frac{r * 3}{360}\right)$
10/1 - 10/4	Weekdays	$\left(1 + \frac{r * 1}{360}\right)^4$
10/5 - 10/8	Weekend Days (incl. Columbus Day)	$\left(1 + \frac{r * 4}{360}\right)$
10/9 - 10/11	Weekdays	$\left(1 + \frac{r * 1}{360}\right)^3$
10/12 - 10/14	Weekend Days	$\left(1 + \frac{r * 3}{360}\right)$
10/15 - 10/16	Weekdays	$\left(1 + \frac{r * 1}{360}\right)^2$

We then employ Newton's method to calculate an average expected $r \approx 0.02165$. This implies that the market has priced in a forward rate of approximately 2.165% after the Federal Reserve's upcoming meeting.



Having solved for the one-month contract, we work our way through the available contract prices and CB meeting dates to solve for all available forward rates.

Finally, we take the data from curve that we constructed and express it in one of two categories: the Current Implied Overnight Rate or the Post-Meeting Implied Overnight Rate. In this example, the Current Implied O/N Rate, the overnight rate expected to prevail up until the Sep. 26th 2018 meeting is ~1.919. You can see the Post-Meeting Implied Overnight Rates, the interest rates that are expected to follow an FOMC meeting, for each meeting in the *Implied Rate* column.



Implied Data

The implied data in WIRP is the data displayed in the data grid and Current Implied O/N Rate data directly above the grid. The implied data consists of five related data items, three interest rate expressions and two estimates of CB movements based on those expressions.

Interest Rate Expressions

In the previous sections, we saw how one can estimate forward rates expected before and after a meeting that would solve out to the individual rates surrounding CB meetings. In the interest rate expressions, we are simply labeling the data in a slightly different manner.

Current Implied Overnight Rate

The forward rate that is estimated before the first (upcoming) meeting is labeled as the 'Current Implied Overnight Rate.' This interest rate becomes the starting point for our other calculations.

Note: The generic tickers for these data do not have meeting-specific counterparts, as there is only one for each model.

Post-Meeting Implied Overnight Rate

The forward rates that are estimated for each CB meeting are labeled as the 'Post-Meeting Implied Overnight Rates.' These are the overnight rates which are expected to prevail after each corresponding CB meeting. A generic ticker references the front meeting, while all other meeting-specific data are referenced via a virtual tail in the MMMYYYY form (e.g., USOAFR **DEC2019** Index).

Implied Overnight Rate Change

The spread between the Current Implied Overnight Rate and the Post-Meeting Implied Overnight Rate. These indicate the total change in the overnight rate expected from today until the conclusion of the corresponding CB meeting.

$$Post\ Meeting\ Imp\ Rate_t - Cur\ Imp\ ON\ Rate = Imp\ Rate\ \Delta$$

A generic ticker references the front meeting, while all other meeting-specific data are referenced via a virtual tail in the MMMYYYY form (e.g., USOACH **DEC2019** Index).

CB Estimates

Once we have the interest rate expressions, we can create estimates of how the CB movement 'priced into' the chosen instrument's curve.

Number of Single Hikes or Cuts

The total, cumulative number of single hikes or cuts expected by the conclusion of a corresponding CB meeting. This is calculated as:

$$\frac{Implied\ Rate\ Change_t}{Assumed\ Rate\ Move} = Number\ of\ Single\ Hikes\ or\ Cuts_t$$

A generic ticker references the front meeting, while all other meeting-specific data are referenced via a virtual tail in the MMMYYYY form (e.g., USOANM **DEC2019** Index).

Percent of a Single Hike or Cut

The marginal percent of a single hike or a cut priced in at a specific CB meeting. This number is not cumulative. Rather than always referencing the Current Implied Overnight Rate, this number looks at the spread of the implied overnight rates immediately preceding and following a specific central bank meeting, expressed as a percentage of a single CB move.

$$\frac{Post\ Meeting\ Imp\ Rate_t - Post\ Meeting\ Imp\ Rate_{t-1}}{Assumed\ Rate\ Move} * 100 = Percent\ Single\ Hike\ or\ Cut$$

A generic ticker references the front meeting, while all other meeting-specific data are referenced via a virtual tail in the MMMYYYY form (e.g., USOAPR **DEC2019** Index).

The following example from the US Futures model will illustrate.

On September 13th 2018, the US Federal Reserve was expected by markets to hike rates at its upcoming meeting on the 26th. However, even though markets had priced in 3.5 more hikes through the end of the next year, the percent of a single hike or cut for the November meeting only had 1.4% of a hike priced in. Looking back on how the hikes were distributed on that day illustrates why. We can see by looking at the *Implied Rate* column that markets were expecting a hike to take rates from ~1.918 to ~2.165 on the 26th. However, markets were only implying a rate of ~2.168 after the November meeting. This is because, at the time, a dogma persisted that the Fed would only move rates at every other ('big')

meeting. Therefore, the percent of a single hike priced for the 11/8/2018 meeting was very small and would have been calculated as:

$$\frac{2.168 - 2.165}{0.25} * 100 = 1.4\%$$

